

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0077

Ironwood Llc 345 kV
September 24, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0077. The Interconnection Customer proposes a 1057.6 MW natural gas combined cycle facility interconnecting at the Ironwood Llc 345 kV substation in the Pennsylvania area. The request was received on October 22, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 68 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$61,755,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sate St-Robison Pak 230kV Line	Richland-Dekab 500kV	118.8	649	6.1
Ironwood Llc-Fayette 345kV	Co Line Road Substation-Ironwood Llc 500kV	116.8	1,244	49.0
Grizzly-Del Cerro 138kV	Fayette-Mercury 345kV	115.8	164	6.8
Jones Chapel 500-230 kV TX	Dekalb-Jones Chapel 500kV	112.8	767	5.7
Robison Park-Russell Fork 230kV line	Dekalb-Jones Chapel 500kV line	105.2	919	7.8
Bluebonnet-Russell Fork 230kV	Dekalb Substation-Jones Chapel 500kV Line	104.9	712	5.7
Dekalb 230-1 kV Transformer	Richland Substation-Dekalb 500kV line	102.7	869	6.6
Moser Generating Sta-Curecanti 345kV	Tombstone Substation-Bixby Hill 500kV Line	100.3	1,413	7.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Robison Park-State St 230kV	Rebuild line with higher-capacity conductor	\$6,453,000	34% (\$2,189,000)
Ironwood Llc-Fayette 345kV	Rebuild line with higher-capacity conductor	\$36,869,000	83% (\$30,463,000)
Grizzly-Del Cerro 138kV line	Rebuild line with higher-capacity conductor	\$4,012,000	100% (\$4,012,000)
Jones Chapel 500-230 kV TX	Replace transformer with higher-rated unit	\$17,084,000	34% (\$5,795,000)
Robison Park-Russell Fork 230kV line	Reconductor line	\$12,788,000	34% (\$4,338,000)
Bluebonnet-Russell Fork 230kV	Reconductor line	\$26,966,000	34% (\$9,148,000)
Dekalb 230/1 kV Autotransformer	Replace transformer with higher-rated unit	\$10,846,000	34% (\$3,680,000)
Moser Generating Sta-Curecanti 345kV	Reconductor line	\$8,935,000	24% (\$2,130,000)
	Total allocated to project		\$61,755,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.